

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$$

$$(1+ax)^n = \binom{n}{0} + \binom{n}{1}ax + \binom{n}{2}a^2x^2 + \dots + \binom{n}{n}a^nx^n$$

$$(1+x^m)^n = \binom{n}{0} + \binom{n}{1}x^m + \binom{n}{2}x^{2m} + \dots + \binom{n}{n}x^{nm}$$

$$\left\{ \begin{array}{l} \frac{1-x^{n+1}}{1-x} = 1+x+x^2+x^3+\dots+x^n \\ \frac{1}{1-x} = 1+x+x^2+\dots = \sum_{\bar{\lambda}=0}^{\infty} x^{\bar{\lambda}} \\ \frac{1}{1-ax} = 1+ax+(ax)^2+\dots = \sum_{\bar{\lambda}=0}^{\infty} a^{\bar{\lambda}} x^{\bar{\lambda}} \end{array} \right.$$

$$\frac{1}{1-x} = 1+x+x^2+\dots = \sum_{\bar{\lambda}=0}^{\infty} x^{\bar{\lambda}}$$

$$\frac{1}{1-ax} = 1+ax+(ax)^2+\dots = \sum_{\bar{\lambda}=0}^{\infty} a^{\bar{\lambda}} x^{\bar{\lambda}}$$

$$\frac{1}{(1+x)^n} = \binom{-n}{0} + \binom{-n}{1}x + \binom{-n}{2}x^2 + \dots$$

$$= \sum_{\bar{\lambda}=0}^{\infty} \binom{-n}{\bar{\lambda}} (-x)^{\bar{\lambda}}$$

$$= 1 + (-1) \binom{n+1-1}{1} x + (-1)^2 \binom{n+2-1}{2} x^2 + \dots$$

$$= \sum_{\bar{\lambda}=0}^{\infty} (-1)^{\bar{\lambda}} \binom{n+\bar{\lambda}-1}{\bar{\lambda}} x^{\bar{\lambda}}$$

$$\frac{1}{(1-x)^n} = \binom{-n}{0} + \binom{-n}{1}(-x) + \binom{-n}{2}(-x)^2 + \dots$$

$$= \sum_{\bar{\lambda}=0}^{\infty} \binom{-n}{\bar{\lambda}} (-x)^{\bar{\lambda}}$$

$$= 1 + (-1) \binom{n+1-1}{1} (-x) + (-1)^2 \binom{n+2-1}{2} (-x)^2 + \dots$$

$$= \sum_{\bar{\lambda}=0}^{\infty} \binom{n+\bar{\lambda}-1}{\bar{\lambda}} x^{\bar{\lambda}}$$